

Inventory Control

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* Inventory

- The resource (or items) kept on stock for future use are called as inventory. It is idle resource, but usable.
- The word 'inventory' means simply a stock of idle resources of any kind having economic value.
 - An inventory system is the set of policies and controls that monitor and evaluate the levels of inventory. It determines:
 - How much to order?
 - When to order?
 - What level of inventory should be maintained?

* Types of Inventory

- Raw materials: Basic materials used for manufacturing goods. E.g. wood used to make furniture.
- Work-in-process (WIP): semi-finished goods that are still in production: e.g. A car on the assembly line before engine fitting.
- Finished goods: products / goods that are completed and ready for sale. e.g. Ready-made clothes in a retail store.

* Inventory costs → Material cost

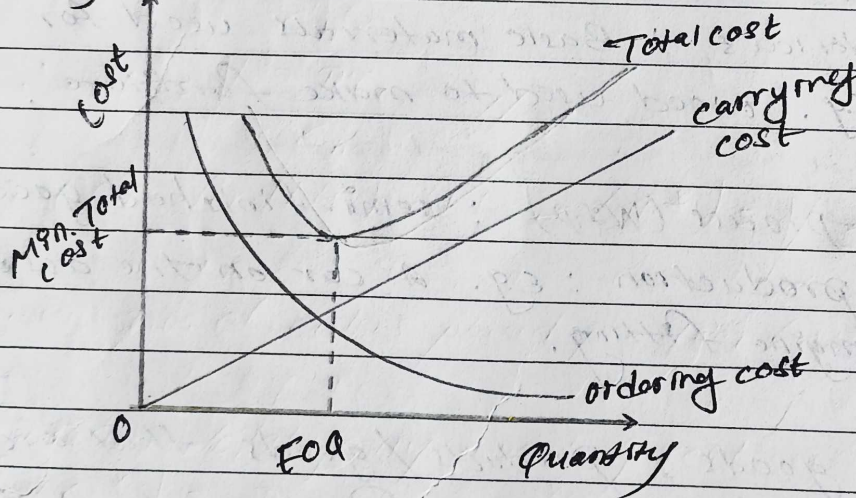
- C_p • purchase (production) cost: The per unit cost of purchasing items from suppliers.
- C_h • carrying (holding) cost: cost of storing and maintaining inventory over time (e.g. rent, insurance, depreciation, spoilage).
- C_s • shortage (stock out) cost: penalty cost for running out of stock. It includes the loss of current profit through the sale of item and the loss of goodwill. It also includes the future loss i.e. loss of the profit through sale of item in the future.

Co • **ordering (setup) cost**: cost incurred every time an order is placed for inventory. E.g. shipping fees, telephone calls, stationary, postage, etc. Frequent ordering can increase these cost, while ordering in bulk may reduce them but increase holding costs.

* Economic order quantity (EOQ)

→ As the size of order increases, the ordering cost decreases while the carrying cost increases.

On the other hand, as the size of order decreases, the ordering cost increases and the carrying cost decreases. These two opposite costs can be shown graphically as:



objective of EOQ:

- Minimize total cost = ordering cost + holding cost

Formula:

$$EOQ = \sqrt{\frac{2DL_0}{C_H}}$$

Where,

- D = Annual demand (units)
- L_0 = ordering cost per order (RS)
- C_H = Holding cost per unit

- Lead time - time gap between placing an order and receiving them.

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* Notation used in models

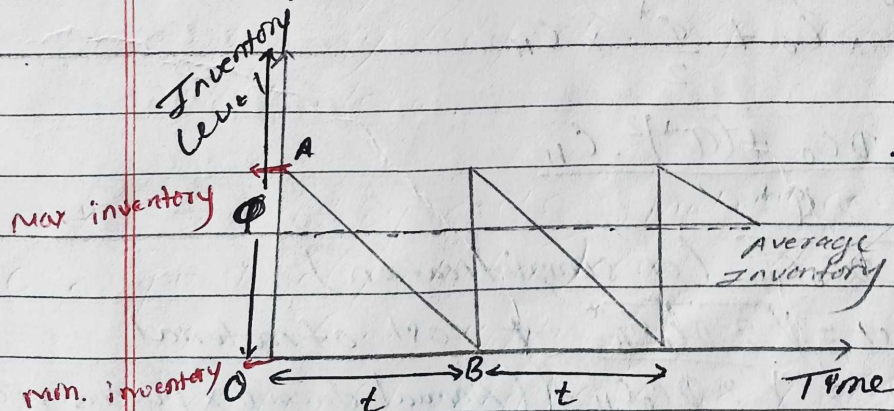
- Q = Order quantity / economic lot size
- D = Annual Demand
- N = NO. of order placed per year
- T_C = Total Inventory cost
- C_o = ordering cost per order
- C = purchase price per unit / cost per unit (Annual average inventory value)
- C_H = carrying / Holding cost per unit
- C_s = Shortage cost per unit

* EOQ model without shortage / EOQ with uniform Demand / Fundamental EOQ problem / EOQ with instantaneous Replenishment

→ The objective of this model is to determine the optimum EOQ in such a manner that total cost is minimized. We make the following assumption for this:

- Demand is constant and known
- Replenishment rate (r_p) is instantaneous ^{→ fill up again}
- Lead time is zero
- Purchase price is constant i.e. discounts are not allowed
- carrying cost & ordering cost are known and constant
- Shortages are not allowed

This situation can be represented graphically as:



→ Rise and fall in inventory level for a particular item over time reflects the periodic cycle of depletion and replenishment. ^{→ EOQ}

Here, • $\text{Average inventory} = \frac{Q + 0}{2} = \frac{Q}{2}$ ($\frac{\text{Max. inventory} + \text{min. inventory}}{2}$)

• $\text{ordering cost} = \text{no. of order per year} \times \text{ordering cost per order}$
 $= \frac{\text{Annual Demand}}{\text{Order quantity}} \times C_o = \frac{D}{Q} \times C_o$

$$\frac{d(x^1)}{dx} = 1 \cdot x^{1-1} = 1$$

(holding)

$$\frac{d(x^n)}{dx} = n \cdot x^{n-1}$$

$$\frac{d(x^{-1})}{dx} = -1 \cdot x^{-1-1} = -\frac{1}{x^2}$$

$$\frac{d(kx)}{dx} = k$$

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- Carrying cost = Average units in inventory $\times$ Carrying cost per unit

$$= \frac{Q}{2} \times C_H$$

- Total inventory cost = (ordering cost + carrying cost)

$$TC = \frac{D}{Q} \times C_0 + \frac{Q}{2} \times C_H$$

Differentiating TC w.r.t Q & then equating it to zero.

$$D C_0 \left(-\frac{1}{Q^2} \right) + \frac{C_H}{2} \times 1 = 0$$

$$\text{or, } \frac{C_H}{2} = \frac{D C_0}{Q^2}$$

$$\text{or, } Q^2 = \frac{2 D C_0}{C_H}$$

$$\therefore Q = \sqrt{\frac{2 D C_0}{C_H}} //$$

Thus, EOQ for this model is $(Q^*) = \sqrt{\frac{2 D C_0}{C_H}}$

Hence, optimum no. of orders placed per year $(N^*) = \frac{D}{Q^*}$

- The minimum total yearly inventory cost is

$$(TC^*) = \frac{D}{Q^*} \times C_0 + \frac{Q^*}{2} \times C_H$$

$$= \frac{2 D C_0 + (Q^*)^2 \cdot C_H}{2 Q^*}$$

$$= \sqrt{2 D C_0 C_H} \text{ (on simplification)}$$

- Total minimum cost = TC^* + cost of material

$$= \sqrt{2 D C_0 C_H} + (\text{Annual demand} \times \text{purchase price})$$

$$= \sqrt{2 D C_0 C_H} + (D \times C)$$

- $\therefore$ Total inventory cost = ordering cost + carrying cost + material cost

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Formula

- $EOQ = \sqrt{\frac{2DC_o}{C_h}}$
- $\text{Max. stock} = EOQ + \text{safety stock}$
- $\text{Min. stock} = 0$ (safety stock)
- $\text{Average inventory} = \frac{EOQ}{2} \text{ i.e. } \left(\frac{\text{Max. stock} + \text{min. stock}}{2} \right)$
- $\text{No. of order per year (N)} = \frac{D}{EOQ}$
- $\text{Carrying cost} = \text{Average inventory} \times \text{carrying cost per unit}$
 $= \frac{EOQ}{2} \times C_h$
- $\text{Ordering cost} = \text{No. of order per year} \times \text{ordering cost per order}$
 $= \frac{D}{EOQ} \times C_o \text{ or } N \times C_o$
- $\text{Material cost} = \text{Annual Demand} \times \text{purchase price per unit}$
 $\text{purchase cost} = D \times C$
- $\text{Total inventory cost (Tc)} = \text{ordering cost} + \text{carrying cost} + \text{Material cost}$

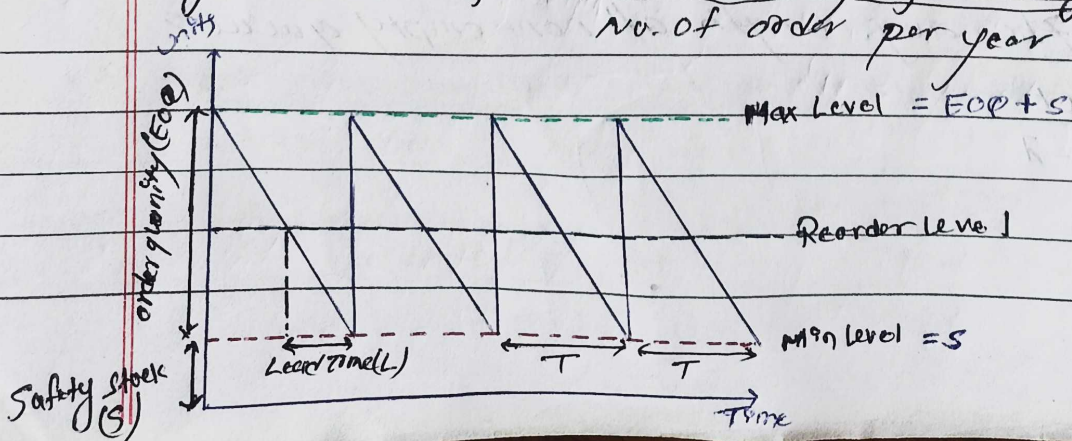
Where,

EOQ = Economic order quantity

D = Annual demand

C = Cost per unit

- $\text{Reorder level (ROL)} = \text{Daily demand} \times \text{Lead time} + \text{safety stock}$ (if given)
 $= \frac{\text{Annual Demand (D)}}{\text{No. of working days (N)}} \times \text{Lead time (L)}$
- $\text{cycle time (T)} = \frac{\text{No. of working days in a year (or } EOQ)}{\text{No. of order per year (D)}}$



- economic lot size = economic order quantity
- purchase price (C) = cost per unit (Annual average inventory value)

* Example

Q. An enterprise requires 1000 units per month. The ordering cost is estimated to be Rs. 50 per order. In addition to Rs. 1, the carrying costs are 5% per unit of average inventory per year. The purchase price is Rs. 20 per unit. Find the economic lot size to be ordered and the total minimum cost.

Soln

Given here,

$$\text{Annual Demand (D)} = \text{monthly demand} \times 12$$

$$= 1000 \times 12$$

$$= 12000 \text{ units/year}$$

$$\text{ordering cost (C}_o\text{)} = \text{Rs. 50 per order}$$

$$\text{purchase price (C)} = \text{Rs. 20 per unit}$$

$$\text{carrying cost (C}_H\text{)} = 1 + 5\% \text{ of } 20$$

$$= \text{Rs. 2 per unit}$$

now,

$$i. \text{ Economic lot size (EOQ)} = \sqrt{\frac{2DC_o}{C_H}}$$

$$= \sqrt{\frac{2 \times 12000 \times 50}{2}}$$

$$= \sqrt{60000} = 774.6 \approx 775 \text{ units}$$

$$ii. \text{ Total minimum cost} = \sqrt{2DC_oC_H} * \text{cost of material}$$

$$= \sqrt{2 \times 12000 \times 50 \times 2} * (12000 \times 20)$$

$$= \text{Rs. 241500}$$

Q.2. The XYZ manufacturing company uses 12000 units of raw material annually, which cost Rs. 1.25 per unit. placing each order costs Rs. 15 and carrying costs are 15% per year of per unit of the average inventory. Find EOQ.

Soln:

Given here,

Annual Demand (D) = 12000 units/year

Purchase cost (C) = 1.25 per unit

Ordering cost (C_o) = Rs. 15

Carrying cost (C_H) = 15% of 1.25
= 0.1875

Now,

$$EOQ = \sqrt{\frac{2DC_o}{C_H}}$$

$$= \sqrt{\frac{2 \times 12000 \times 15}{0.1875}}$$

$$= 1385.64 \approx 1386 \text{ units}$$

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Q3, An auto industry purchases spark plugs at the rate of Rs. 25 per piece. The annual consumption of spark plug is 18000 numbers. If the ordering cost is Rs. 250 per order and carrying cost is 25% p.a., what would be the EOQ? If the supplier of spark plugs offers a discount of 5% for order quantity of 3000 numbers per order, do you accept the discount offer.

Soln

Given here,

Purchase price (C) = Rs. 25 per piece

Annual Demand (D) = 18000

Ordering cost (C_o) = Rs. 250 per order

Carrying cost (C_H) = 25% of Rs. 25
= 6.25

Now.

$$EOQ = \sqrt{\frac{2 D C_o}{C_H}}$$

$$= \sqrt{\frac{2 \times 18000 \times 250}{6.25}}$$

$$= 1200 \text{ units}$$

Again, comparing total costs
1st case (without discount)

We have, $EOQ = 1200$

- No. of order per year (N) = $\frac{\text{Annual Demand}}{EOQ} = \frac{18000}{1200} = 15$
- ordering cost = no. of order per year $\times$ ordering cost per order
 $= N \times C_o$
 $= 15 \times 250 = \text{Rs. } 3750$
- carrying cost = $\frac{\text{Average inventory} \times \text{carrying cost per unit}}{2}$
 $= \frac{EOQ \times C_H}{2} = \frac{1200 \times 6.25}{2} = \text{Rs. } 3750$
- material cost = Annual Demand $\times$ purchase price per unit
 (purchase cost) $= 18000 \times 25 = \text{Rs. } 4,50,000$
- $\therefore$ Total cost (TC) = ordering cost + carrying cost + material cost
 $= 3750 + 3750 + 4,50,000$
 $= \text{Rs. } 4,57,500 //$

2nd case (with discount)

We have, $EOQ = 3000$, discounted unit cost (C) = $25 - 5\% \text{ of } 25$
 $= \text{Rs. } 23.75$

- No. of order per year (N) = $18000 / 3000 = 6$
- ordering cost = $N \times C_o = 6 \times 250 = \text{Rs. } 1500$
- carrying cost = $\frac{EOQ \times C_H}{2} = \frac{3000 \times (0.25 \times 23.75)}{2} = \text{Rs. } 8906.25$
- purchase cost = $D \times C = 18000 \times 23.75 = \text{Rs. } 4,27,500$
- $\therefore$ Total cost = $1500 + 8906.25 + 4,27,500 = \text{Rs. } 4,37,906.25$
- $\Rightarrow$ Yes, I will accept the discount offer, as it lowers the total annual cost by 19,593.75 rupees.

Q. 4, A company uses annually 50,000 units of an item costing Rs. 1.20. It operates 250 days in a year and procurement time is 10 days. Each order cost Rs. 45 and inventory carrying cost is 15% of the annual average inventory value. If the safety stock is 500 units, find EOQ, Re-order level, minimum, maximum & average inventory levels.

Soln

Given here,

$$\text{Annual Demand (D)} = 50,000 \text{ units}$$

$$\text{cost per unit (C)} = \text{Rs. } 1.20$$

$$\text{ordering cost (C}_o\text{)} = \text{Rs. } 45 \text{ per order}$$

$$\text{Carrying cost (C}_h\text{)} = 15\% \text{ of } 1.20$$

$$= 0.18$$

$$\text{Safety stock (S)} = 500 \text{ units}$$

$$\text{No. of working days (N)} = 250 \text{ days, Lead Time (L)} = 10 \text{ days}$$

now,

$$\text{EOQ} = \sqrt{\frac{2DC_o}{C_h}}$$

$$= \sqrt{\frac{2 \times 50000 \times 45}{0.18}}$$

$$= 5000 \text{ units}$$

$$\text{Reorder Level (ROL)} = \frac{D}{N} \times L + S = \frac{50000}{250} \times 10 + \text{Safety stock}$$

$$= 2000 + 500$$

$$= 2500 \text{ units}$$

$$\text{Minimum Inventory} = \text{Safety stock} = 500 \text{ units}$$

$$\text{Maximum Inventory} = \text{EOQ} + \text{Safety stock}$$

$$= 5000 + 500$$

$$= 5500 \text{ units}$$

$$\text{Average inventory} = \frac{\text{Max. Inventory} + \text{Min. inventory}}{2}$$

$$= \frac{5500 + 500}{2} = 3000 \text{ units}$$

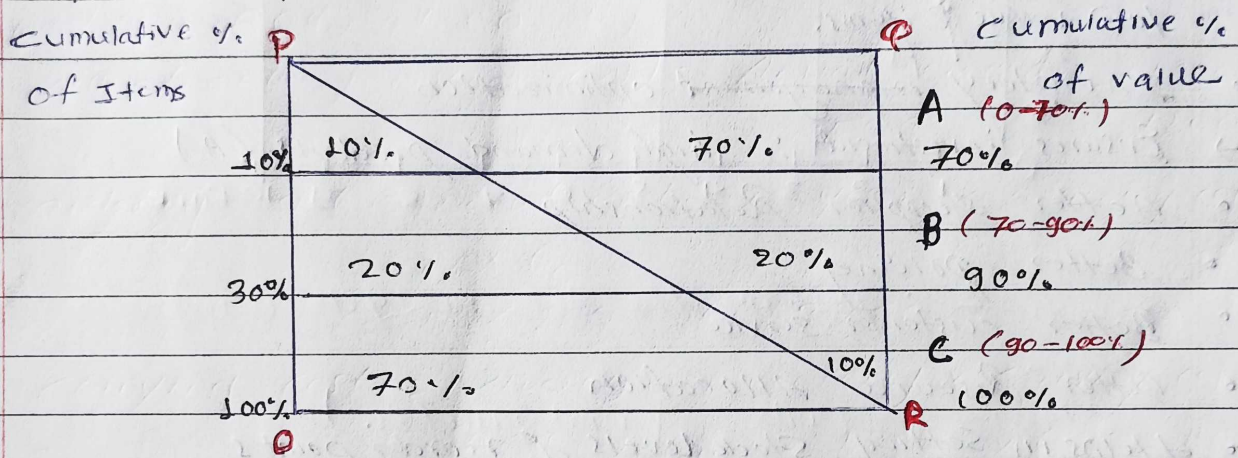
* ABC Analysis / ABC Inventory System

→ ABC stands for Always Better Control in inventory. It refers to the method that categorize inventory items based on their value and importance to a business.

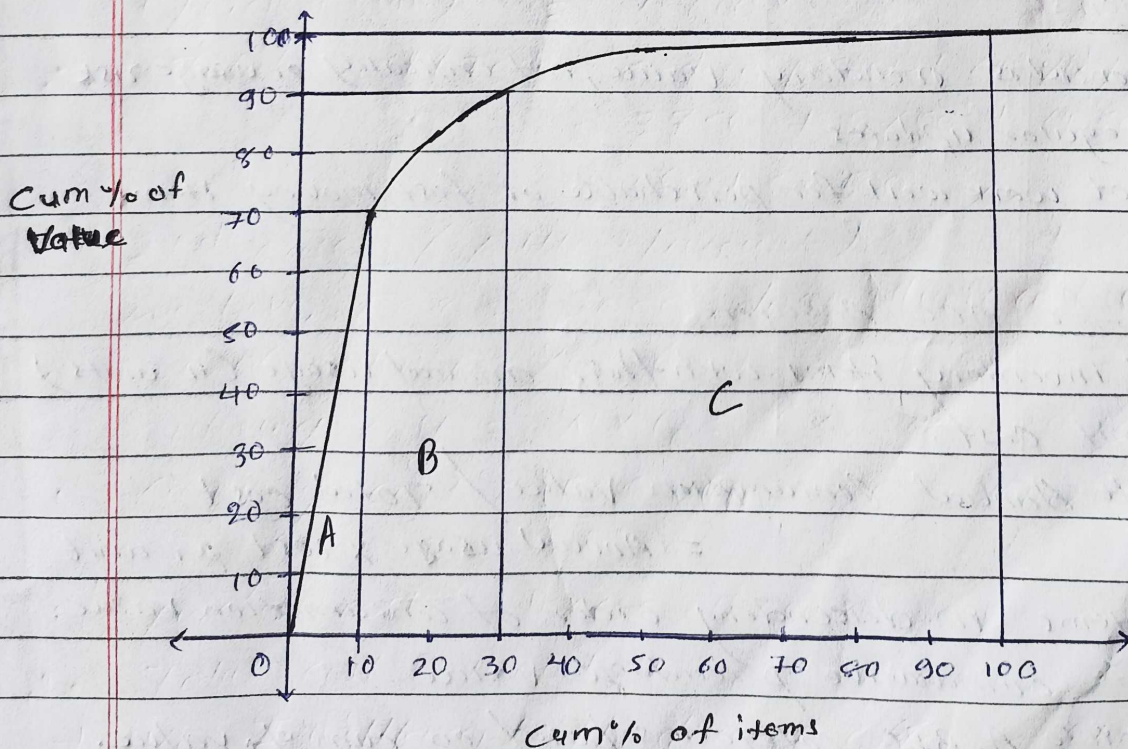
It divides inventory into 3 classes:

- | | Items | Value |
|--|-------|-------|
| • A - Few items with large value | 10% | 80% |
| • B - Moderate items with moderate value | 20% | 20% |
| • C - More items with small value | 70% | 10% |

This is represented in the figure below



Here, upper ΔPQR refers to value of inventory & lower ΔPQR refers to items of inventory



This categorization of inventory helps business prioritize inventory management efforts, focusing more closely on high-value items while potentially streamlining processes for lower value items.

* Benefits of ABC Analysis

- Improved Inventory management
 - focuses efforts on high value A items reducing time wasted on low value items.
- Inventory management optimization
 - Ensures investment in high demand products (A)
- Better supplier Relationship
- Better pricing
- Better customer service
- Better Resource Allocation
- Helps in setting stock levels & reorder points
- Reduces excess inventory and carrying cost

* Limitations

- only considers monetary value, not critically or usage risk
- Needs regular updates
- may not work well for perishable or fast moving items

* Steps for ABC Analysis

- 1) List all inventory items with their annual usage (in units) and unit cost.
- 2) Calculate Annual Consumption Value / Total cost

$$= \text{Annual usage} \times \text{cost per unit}$$
- 3) Rank items in descending order of consumption value.
- 4) Calculate cumulative percentage value
- 5) Classify into A, B & C categories based on cum % value.

* Example: 1

From the information given below, draw an ABC inventory chart.

Items	1	2	3	4	5	6	7	8	9	10
Annual usage in units	1500	5000	1800	2300	800	7000	400	10000	8000	1500
Unit cost (Rs.)	5	4	3	1	5	10	8	8	7	3

Soln:

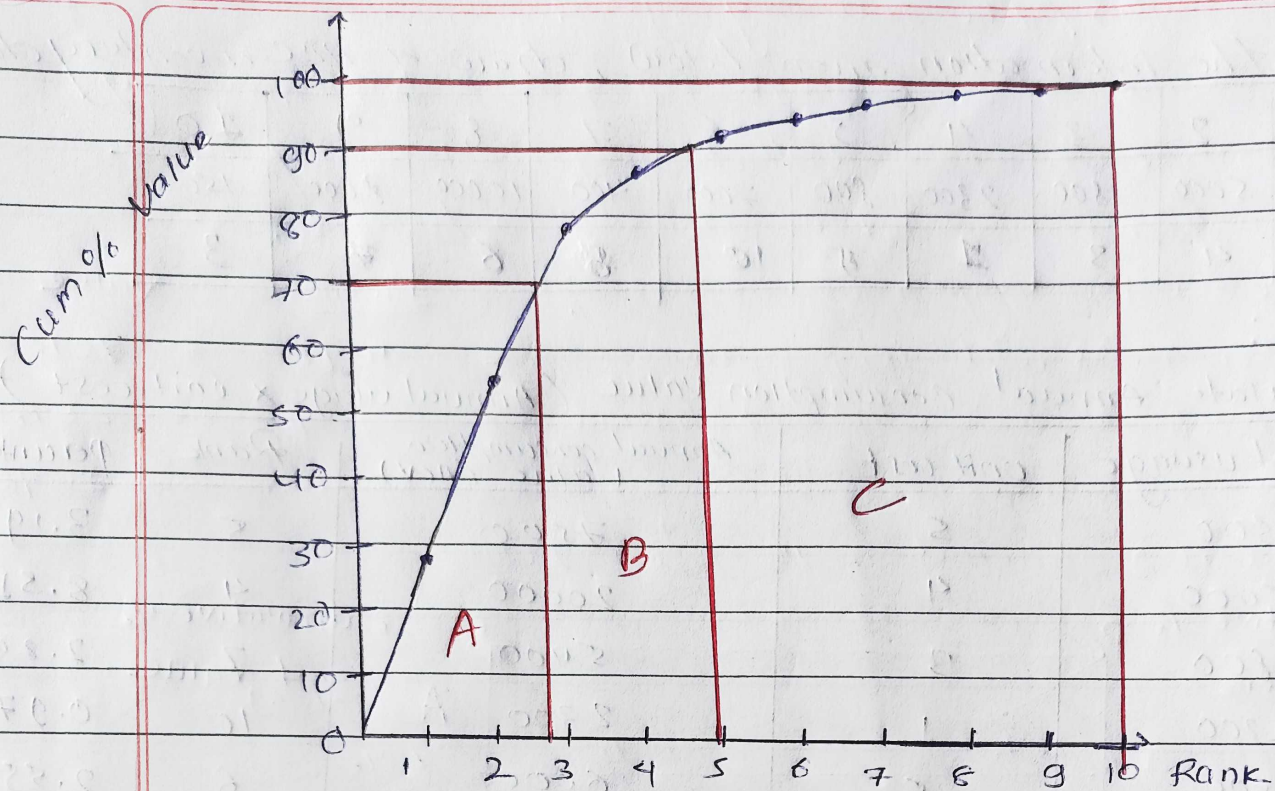
Calculate Annual consumption value (Annual usage $\times$ unit cost)

Items	Annual usage	unit cost	Annual consumption Value (ACV)	Rank	percentage
1	1500	5	7500	5	3.19
2	5000	4	20000	4	8.51
3	1800	3	5400	7	2.29
4	2300	1	2300	10	0.97
5	800	5	6000	6	2.55
6	7000	10	70000	1	29.79
7	400	8	3200	9	1.36
8	10000	8	80000	2	25.54
9	8000	7	56000	3	23.83
10	1500	3	4500	8	1.91

Total = 2,34,900

List the items in descending order based on Annual consumption.

Rank	ACV	percentage	Cumulative % value	Item
1	70000	29.79	29.79	6
2	60000	25.54	55.33	8
3	56000	23.83	79.16	9
4	20000	8.51	87.67	2
5	7500	3.19	90.86	1
6	6000	2.55	93.41	5
7	5400	2.29	95.7	3
8	4500	1.91	97.61	10
9	3200	1.36	98.97	7
10	2300	0.97	99.94	4



* Example 2 : Perform the ABC analysis using the following data.

Item	Units	Unit Price (Rs.)
1	700	50
2	2400	30
3	150	100
4	60	220
5	3800	15
6	4000	5
7	6000	2
8	200	35
9	3000	80
10	2900	4

Soln :

Items	Units	Unit price	Total cost	Percentage	Rank
1	700	50	35000	14.04	3
2	2400	30	72000	28.88	1
3	150	100	15000	6.02	5
4	60	220	13200	5.29	6
5	3800	15	57000	22.86	2
6	4000	5	20000	8.02	4
7	6000	2	12600	5.05	7
8	300	35	10500	4.21	9
9	30	80	2400	0.96	10
10	2900	4	11600	4.65	8

Total = 249300

Rank	Items	percentage	Cum. percentage value
1	2	28.88	28.88
2	5	22.86	51.74
3	1	14.04	65.78
4	6	8.02	73.81
5	3	6.02	79.82
6	4	5.29	85.12
7	7	5.05	90.17
8	10	4.65	94.82
9	8	4.21	99.04
10	9	0.96	100

